

ESA MODEL QUESTION PAPER

M TECH DATA SCIENCE AND ARTIFICIAL INTELLIGENCE_SEMESTER 2

Subject: Generative AI and LLM

Duration: 3 hours | Total Marks: 100

Instructions:

Answer all questions.

Write clear, concise answers with proper reasoning for scenario-based questions.

Q1.

1a (i) Explain the fundamental difference between Discriminative and Generative models with examples. What makes a Generative model suitable for tasks like text generation? [3 marks]

(ii) Describe the role of the Transformer architecture in modern LLMs. Why is the self-attention mechanism critical for LLMs like GPT-4 and LLaMA? [4 marks]

1b. Identify suitable models for each of the following modalities:

Text-to-Text, Image-to-Vector , Audio-to-Text [6 marks]

1c. You are designing a chatbot for customer support. The initial prompts are vague, leading to irrelevant responses. Demonstrate with examples how you would improve this using:

(i) Zero-shot prompting with better instruction clarity [3 marks]

(ii) Few-shot learning with 2-3 examples to guide the model [3 marks]. What trade-off do you observe?

1d. Scenario: Your organization is choosing between API-based LLMs and self-hosted. Discuss the latency, scalability, and update strategy implications for:

(i) A real-time customer-support chatbot that generates responses to customer queries [3 marks]

(ii) A batch-processing pipeline that summarizes a large collection of business documents [3 marks].

Q2.

2a. (i) Explain what Parameter-Efficient Fine-Tuning (PEFT) solves compared to traditional full fine-tuning. Why is this critical for tuning large models like LLaMA? [3 marks]

(ii) Explain the principle behind LoRA. Why does LoRA train small adapter matrices instead of updating all model weights? [2 Marks]

2b. (i) What is QLoRA and how does it differ from standard LoRA? [3 marks]

(ii) Under what scenarios would you choose QLoRA over LoRA, and what quantization techniques does QLoRA employ? [4 marks]

2c. Scenario: You are planning to fine-tune a 13B-parameter LLaMA model using LoRA for an instruction-following task. You have two constraints: (i) limited GPU memory (16 GB VRAM), and (ii) faster training convergence. Your initial hyperparameters are `learning_rate = 1e-4`, `rank = 16`, `alpha = 32`, and `epochs = 3`.

Suggest three appropriate hyperparameter adjustments. For each adjustment, specify the new value and explain how it helps address either the GPU-memory constraint or faster convergence.

[7 marks]

2d. Explain the workflow: Supervised Fine-Tuning (SFT) → Reward Modeling → PPO (Reinforcement Learning). Why is alignment through this multi-stage process necessary for models like ChatGPT and Claude, and what role do human preferences play? [6 marks]

Q3.

3a. (i) Explain what embeddings are and how they transform text into vector representations suitable for retrieval. [3 marks]

(ii) Why is semantic similarity (via embeddings) preferable to keyword-based retrieval in a Retrieval-Augmented Generation (RAG) pipeline? [4 marks]

3b. What is HNSW (Hierarchical Navigable Small World) in a vector database? Briefly explain how HNSW-based search differs from general/exhaustive vector search in terms of how the relevant vectors are identified. [6 marks]. [6 marks]

3c. Scenario: You are building an end-to-end RAG pipeline for a medical knowledge base. A user asks: “What are the contraindications for Aspirin?”

(i) Outline the main steps from the user query to the final LLM response, including document chunking, embedding, retrieval, and answer generation. [4 marks]

(ii) Identify one possible failure in the retrieval stage and suggest one suitable solution. [3 marks]

3d. 3(d). Explain the role of chunking in a RAG system. Compare the “one vector per document” and “multiple vectors per document” approaches, and explain why chunking generally improves retrieval quality. [5 marks]

Q4.

4a. (i) Define an AI agent and explain how it differs from a traditional LLM. [3 marks]

(ii) Explain the roles of the Planner, Executor, Memory, and Feedback Loop in an Agentic AI system. [4 marks]

4b. An agent receives the query: “Find the top 3 research papers on Graph Neural Networks and summarize them.”

Explain how the Planner and Executor would work together to accomplish this task. Identify the sequence of actions the agent should perform and explain how the feedback loop can help if the retrieved results are poor. [6 marks]

4c. (i) An agent receives the query: “What is the current population of Japan and calculate 10% of it.”

Using the ReAct framework, write the sequence of Thought → Action → Observation steps that the agent would follow before producing the final answer. [4 marks]

(ii) Consider the following agent workflow: User Query → Agent → Search Tool → Result → Final Answer

The search result is irrelevant to the user's query. Identify the missing step in the workflow and explain how the agent should handle the situation. [3 marks]

4d. A customer-support agent is given access to a database, email system, and external API. The agent can independently select and execute tools.

(i) Identify two risks that could occur when the agent operates autonomously. [2 marks]

(ii) For each risk, suggest an appropriate guardrail or control mechanism. [3 marks]